

1. Requirements & Install

Component	Minimum	Notes
OS	Ubuntu 22.04/24.04 (X11)	Windows/macOS untested here
CPU / RAM	4 cores / 8 GB	60 fps canvas + Kalman filter (CV) inside budget
Rust / Node / Python	>= 1.77 / 20 / 3.10	rustup + npm + sidecar venv
Hardware	optional	3x RTL-SDR + AWOS; synthetic feed without

```
npm install
cd python-sidecar && python3 -m venv venv && venv/bin/pip install duckdb pyserial pytest && cd
..
./scripts/verify.sh
```

1.1 Webkit headers without sudo

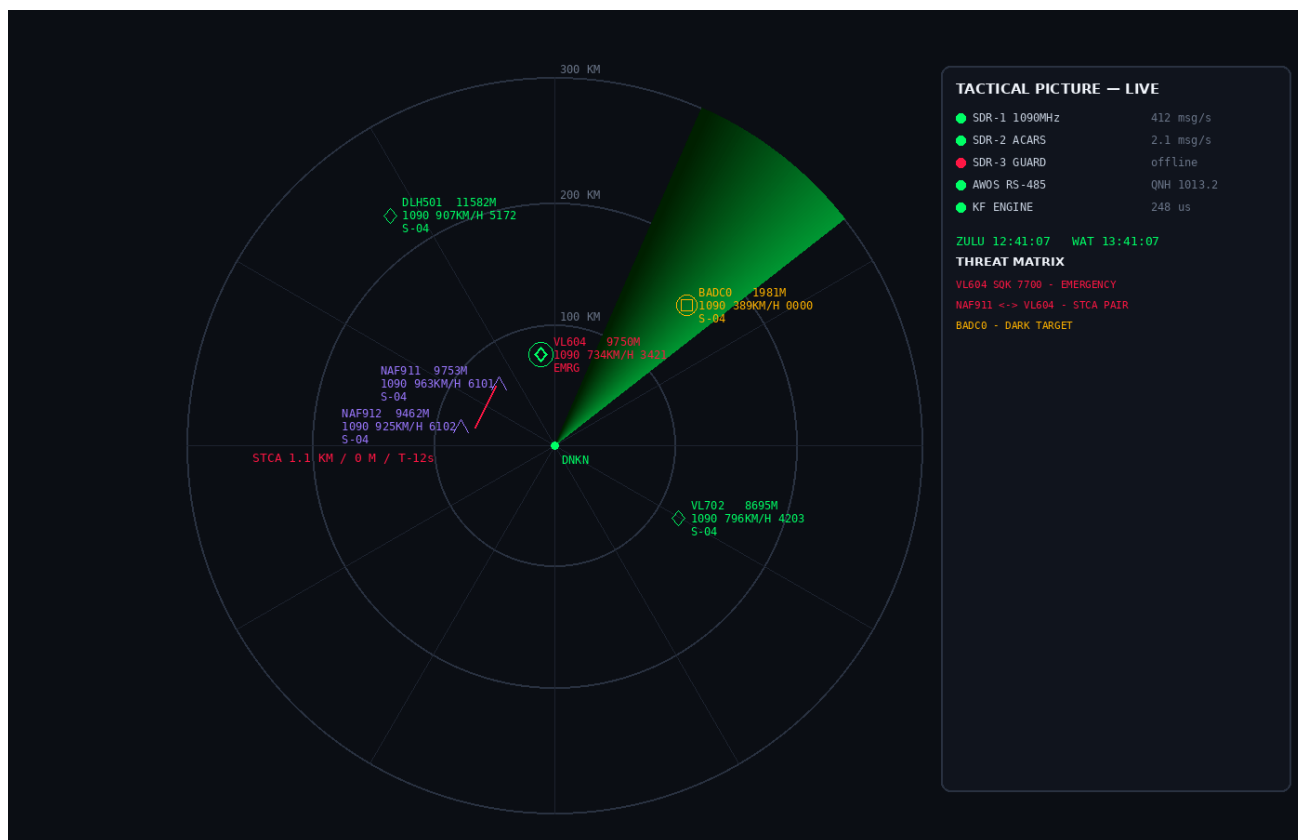
The desktop shell compiles against webkit2gtk-4.1 development headers; the runtime library ships with Ubuntu. `scripts/webkit-env.sh` prepares a local prefix by downloading the -dev packages with `apt-get download` and extracting them via `dpkg -x` into `~/ap-deps/merged`. With `sudo`, a plain `apt install` of `libwebkit2gtk-4.1-dev` and `libsoup-3.0-dev` replaces the whole dance.

2. Launch Modes

```
scripts/desktop.sh      # full application (both windows + runtime + sidecar)
cargo run --manifest-path src-tauri/Cargo.toml --no-default-features \
  --bin aeropulse-headless # terminal tactical console
npm run dev              # browser UI (synthetic feed) -> http://localhost:1420
cd python-sidecar && venv/bin/python main.py --db test.duckdb
```

First desktop launch compiles the dependency tree once; later starts take roughly fifteen seconds. The verification battery `./scripts/verify.sh` runs every suite in one command.

3. Screen Reference



3.1 Master command bar (Screen 0)

- Channel health chips for SDR-1/2/3, AWOS and the KF engine (Kalman filter) — green nominal, flashing red fault.
- Range-scale selector in kilometres (25-450 km, SI policy) with rings at 100/200/300 km.
- ZULU and WAT (UTC+1, no DST) clocks; EMERG 7700 override arms fleet-wide emergency for 30 s.

3.2 Display 1 — tactical radar

Rotating sweep with phosphor fade; MIL-STD-2525D symbols (diamond civil, caret military, quadrangle unknown/anomaly); three-line Flight Data Blocks in SI (altitude metres, speed km/h, source and squawk); dead-reckoning halos; flashing red STCA connectors with minima labels; geofence polygons; R&B ruler tool (click anchor, move, double-click clears); intercept calculator panel.

3.3 Display 2 — operations HUD

Flight-strip bay (arrivals/departures with green/amber/red edge states), triple-fusion weather panel (AWOS surface block, upper-air matrix with provenance, dust-layer top, raw D-ATIS/METAR feed), threat matrix, and diagnostics with the spectrum plot (simulated: message-rate synthesis until RF tap) and KF latency (Kalman filter) gauge against the 2 ms ceiling.

4. Operational Walkthrough

- Boot: six tracks already filtered (Kalman filter converges in ~12 fixes); weather fills within the first ACARS/AWOS cadence.
- T+45 s: VL604 squawks 7700 — symbol, FDB annotation and strip edge go red; threat matrix logs the emergency.
- T+60-80 s: NAF911 crosses VL604's path — red dashed STCA connector flashes at 2 Hz with minima label.
- T+112 s: VL604 drops off RF — dashed coasting halo, FDB source flips to DR, leader line keeps projecting.
- Rolling: dark target flagged after its arming window; ACARS METAR every 45 s; AWOS every 30 s.

5. Data & Persistence

The sidecar writes `aeropulse.duckdb` (`track_positions`, `stca_alerts`, `weather_observations`) with transactional batches every 2 s. Query example:

```
python-sidecar/venv/bin/python - <<'PY'
import duckdb
c = duckdb.connect('python-sidecar/aeropulse.duckdb', read_only=True)
print(c.execute('SELECT callsign, count(*) FROM track_positions GROUP BY 1').fetchall())
PY
```

6. Troubleshooting

Symptom	Cause	Fix
webkit pkg-config not found	dev headers absent	scripts/webkit-env.sh or apt install
cargo run: which binary	older checkout	git pull; default-run is set in Cargo.toml
Vite config-loader warning	pre-type:module layout	fixed; update checkout
No weather data	sidecar missing	check [sidecar] log lines; persistence degrades gracefully
STCA never fires in browser	demo window timing	conflict window t=20-140 s after load; reload